ANSWER KEY & MARKING SCHEME SET A

Chapter 5: **Life Processes** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** chloroplast	Contains chlorophyll
2	**(b)** liver	Stored in gall bladder
3	**(d)** muscle cells during lack of oxygen	Causes cramps
4	**(b)** Incorrect — the pulmonary artery carries deoxygenated blood	Exception
5	**(a)** Student A is correct	Turgidity of guard cells
6	**(b)** nephron	Filtration unit
7	**(a)** pulmonary vein	Oxygenated
8	**(a)** salivary amylase	Starch → sugar
9	**(c)** phloem	Translocation
10	**(b)** 4	Two atria, two ventricles

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Ventricles pump blood to various organs (and lungs) with greater force.	2
12	$6\,CO_2 + 12\,H_2O \xrightarrow[\text{chlorophyll}]{\text{sunlight}} C_6H_{12}O_6 + 6\,O_2 + 6\,H_2O$	2
13	Ethanol and carbon dioxide (with energy)	1+1
14	True — herbivores eating grass need a longer small intestine to digest cellulose.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) makes medium acidic, activates pepsin, kills germs (ii) digests proteins (iii) protects the inner lining from acid	1+1+1
16	Bowman's capsule, glomerulus, tubule, collecting duct; filtration in glomerulus, selective reabsorption of glucose, amino acids, salts, water in tubule → urine	1½+1½
17	Blood passes twice through the heart in one cycle — pulmonary (heart–lungs–heart) and systemic (heart–body–heart); keeps oxygenated and deoxygenated blood separate for efficient oxygen supply to warm-blooded animals	2+1
	OR: Aerobic: needs oxygen, complete breakdown to CO_2 + H_2O, more energy (in mitochondria); anaerobic: no oxygen, partial breakdown to ethanol/lactic acid, less energy (in cytoplasm)	3
18	Through xylem; root cells take in ions creating a pressure difference; transpiration pull — evaporation from leaves draws water up; also regulates temperature	3
	OR: CO_2 — from air through stomata; water — from soil through roots; sunlight — absorbed by chlorophyll; minerals from soil	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) receives bile (emulsifies fats) and pancreatic juice; intestinal juice completes digestion: proteins → amino acids, carbohydrates → glucose, fats → fatty acids + glycerol (ii) finger-like projections that increase surface area for absorption (iii) trypsin, lipase (amylase)	2+2+1
	OR: (i) nostrils → pharynx → larynx → trachea (with cartilage rings) → bronchi → bronchioles → alveoli; diaphragm (ii) alveoli give a huge surface (~80 m²), thin walls, rich blood capillaries	3+2
20	(i) to remove existing starch (ii) covered: no blue-black colour; uncovered: blue-black (iii) light is necessary for photosynthesis (iv) to remove chlorophyll	1+2+1+1
	OR: (i) four chambers; right side receives deoxygenated blood from vena cava and pumps to lungs; left side receives oxygenated blood from pulmonary veins and pumps through aorta; valves prevent backflow (ii) blood passes once through the heart; oxygenated at gills	4+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 5: **Life Processes** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** stomata	Guard cells
2	**(a)** green plant	Makes own food
3	**(b)** cytoplasm	Glycolysis
4	**(a)** Correct	Plug leaks
5	**(b)** Student A is correct	Lacteals
6	**(a)** red blood cells	Carries O_2
7	**(a)** cuscuta	Amarbel
8	**(b)** ATP	Adenosine triphosphate
9	**(b)** sphincter muscle	Pyloric sphincter
10	**(d)** ureters	Pair of tubes

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	By forming pseudopodia that engulf food into a food vacuole, where it is digested.	2
12	Xylem (water and minerals) and phloem (food)	1+1
13	The amount of dissolved oxygen in water is much lower than in air, so they breathe faster.	2
14	False — the right side receives and pumps deoxygenated blood to the lungs.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Acid: activates pepsin, kills bacteria; liver: secretes bile which neutralises acid and emulsifies fats	1½+1½
16	When guard cells take up water they swell and the pore opens; when they lose water they shrink and the pore closes.	3
17	In the small intestine: bile salts emulsify fats into small globules; lipase from pancreatic juice converts them into fatty acids and glycerol.	3
	OR: Transport of soluble food via phloem; supplies food to all parts including storage organs, fruits; uses ATP energy to load sucrose, osmotic pressure moves it	1+1+1
18	Artificial kidney: blood passes through tubes in a dialysing fluid; nitrogenous wastes diffuse out; used when kidneys fail.	3
	OR: O_2/CO_2 by diffusion via stomata; excess water by transpiration; wastes stored in leaves that fall, in vacuoles, as resins/gums in old xylem	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) two kidneys, ureters, urinary bladder, urethra (ii) filtration in glomerulus into Bowman's capsule; selective reabsorption of glucose, amino acids, salts, water in tubules (iii) depends on excess water and dissolved wastes; hormones regulate reabsorption	2+2+1
	OR: (i) glucose → pyruvate (cytoplasm); without O_2: → ethanol + CO_2 (yeast); lack of O_2: → lactic acid (muscles); with O_2: → CO_2 + H_2O + energy (mitochondria) (ii) cytoplasm, mitochondria	4+1
20	(i) lymph is colourless, has less protein, no RBCs (ii) carries digested fats from intestine; drains excess fluid back to blood; immunity (iii) plasma, RBC, WBC, platelets (iv) blood clotting	1+2+1+1
	OR: (i) mouth → oesophagus → stomach → small intestine → large intestine → anus; glands: salivary, liver, pancreas (ii) rhythmic contraction of the muscles that pushes food forward (iii) absorbs excess water	3+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.